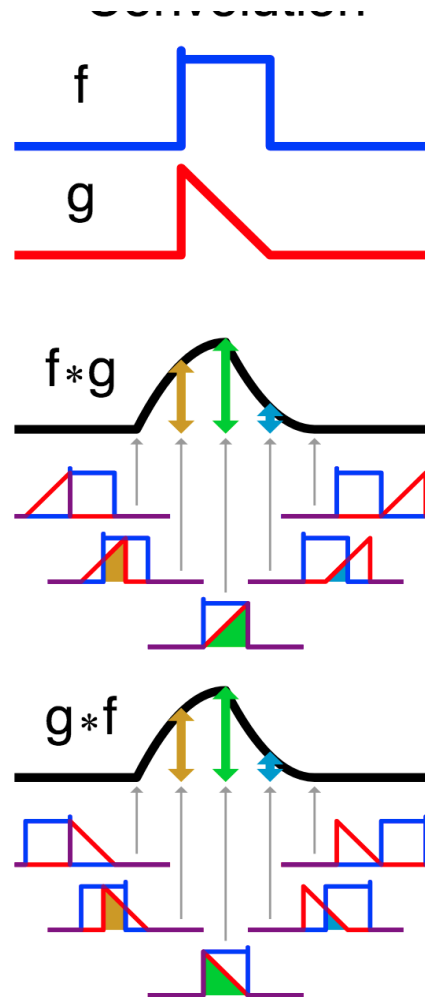




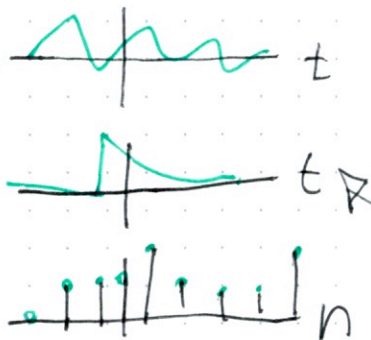
SYDE252 - W03A - Convolution

Dr. Matt Borland

Where we're going...

WHERE WERE GOING:

SIGNALS



SYSTEMS



LTI
INVARIANT

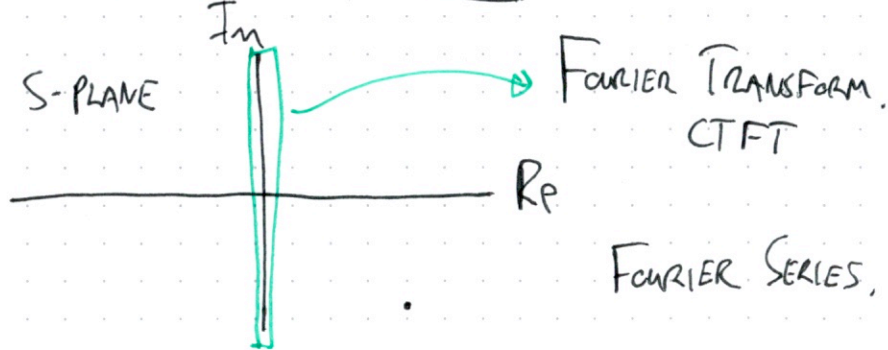
ANALYSE THESE

TOOLS

$\int$
 $\sum$
 $e^{j\omega t}$
 $*$

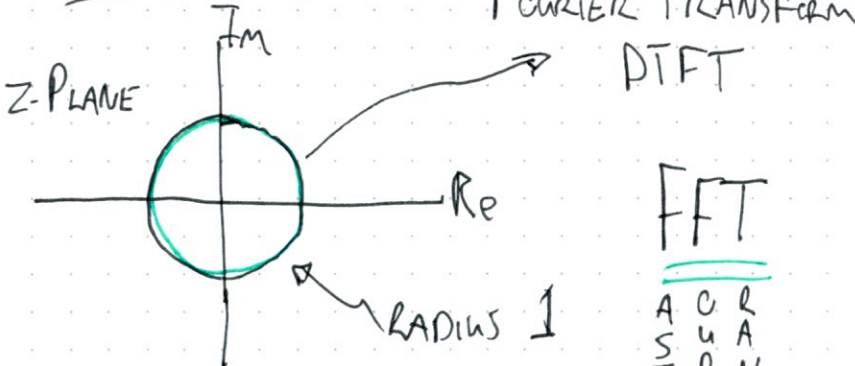
CT

LAPLACE TRANSFORM



DT

Z-TRANSFORM



CONTROLS
SIGNAL PROCESSING.

FFT
ACCELERATOR TRANSFORM

How to get there...

Get the textbook.

Read the chapters in advance of the class. There's a lot of content, and you need to read the book to learn it!

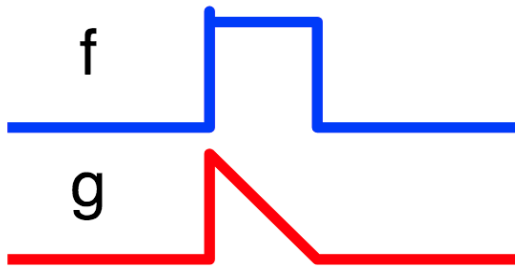
Come to class and pay attention, try the sample problems, engage.

Attempt the assignments before the tutorials, go to the tutorials, then check the solutions after they're posted.

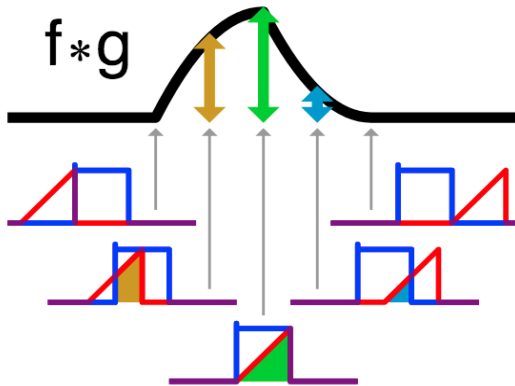
Get help. From me. From the TAs.

Get the textbook.

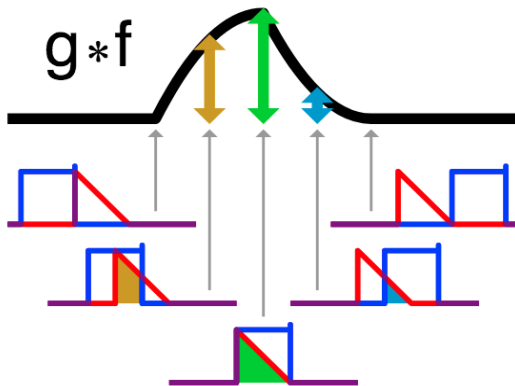
Convolution



$$f(t) * g(t)$$

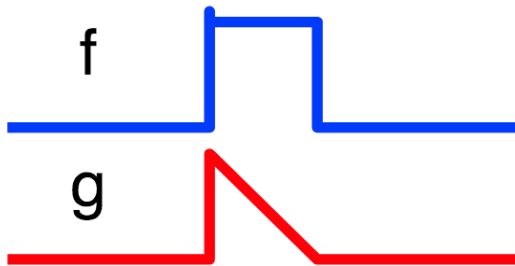


$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$$

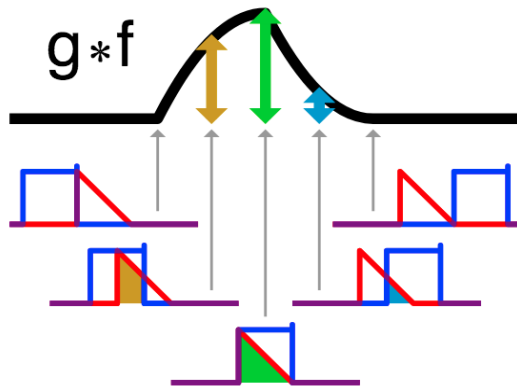
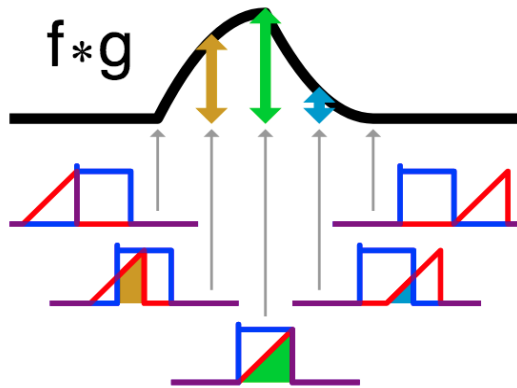


<https://en.wikipedia.org/wiki/Convolution>

Convolution



$$f(t) * g(t)$$



Take: $g(-t)$ ↖ The flip is important!

...and shuffle it along $f(t)$

...and summate the area under the product curve.

<https://en.wikipedia.org/wiki/Convolution>

CT Convolution Properties

Commutative Property

$$x_1(t) * x_2(t) = x_2(t) * x_1(t)$$

Distributive Property

$$x_1(t) * [x_2(t) + x_3(t)] = x_1(t) * x_2(t) + x_1(t) * x_3(t)$$

Associative Property

$$x_1(t) * [x_2(t) * x_3(t)] = [x_1(t) * x_2(t)] * x_3(t)$$

Shift Property

$$\text{If } x_1(t) * x_2(t) = c(t)$$

$$\text{Then } x_1(t) * x_2(t - T) = x_1(t - T) * x_2(t) = c(t - T)$$

CT Convolution

Just solve that integral (fun, right ;))

Luckily we have tables for common functions and convolution pairs.

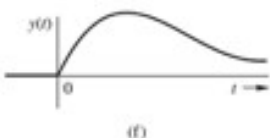
$$x(t) = e^{-t}u(t), \quad h(t) = e^{-2t}u(t)$$

From table 2.1 (pg. 176 in 3rd Edition)

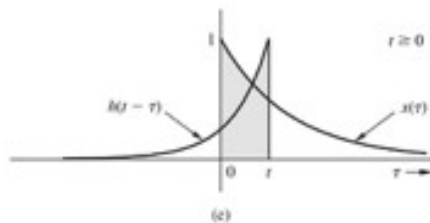
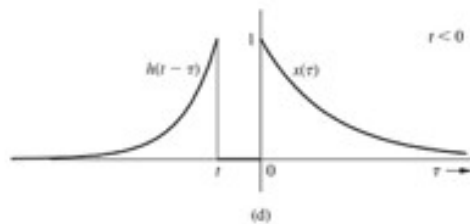
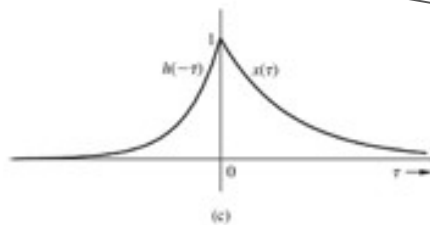
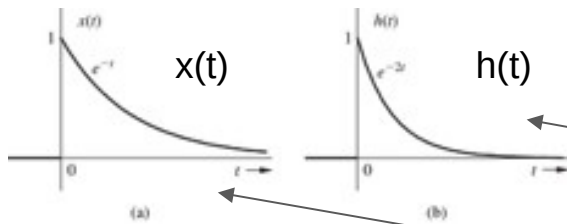
$$\text{For: } x_1(t) = e^{\lambda_1 t}u(t)$$

$$x_2(t) = e^{\lambda_2 t}u(t)$$

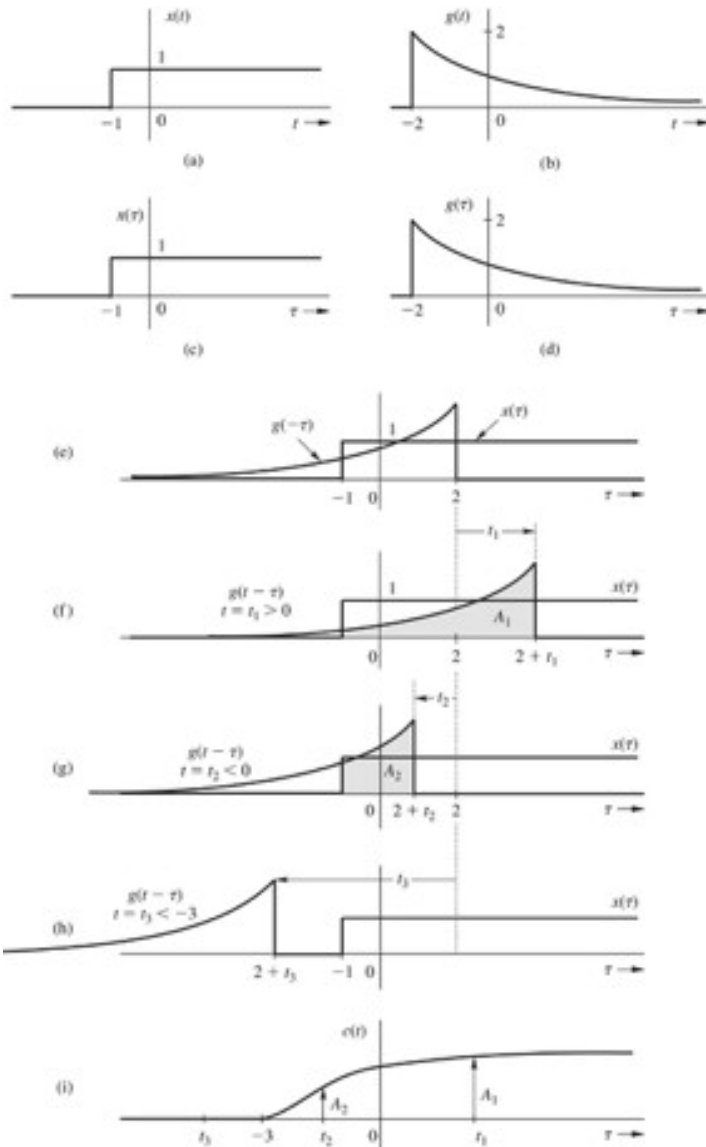
$$x_1(t) * x_2(t) = \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2} u(t)$$

$$x(t) * h(t) = y(t)$$


Check it with W03A_CT_Convolution.m!



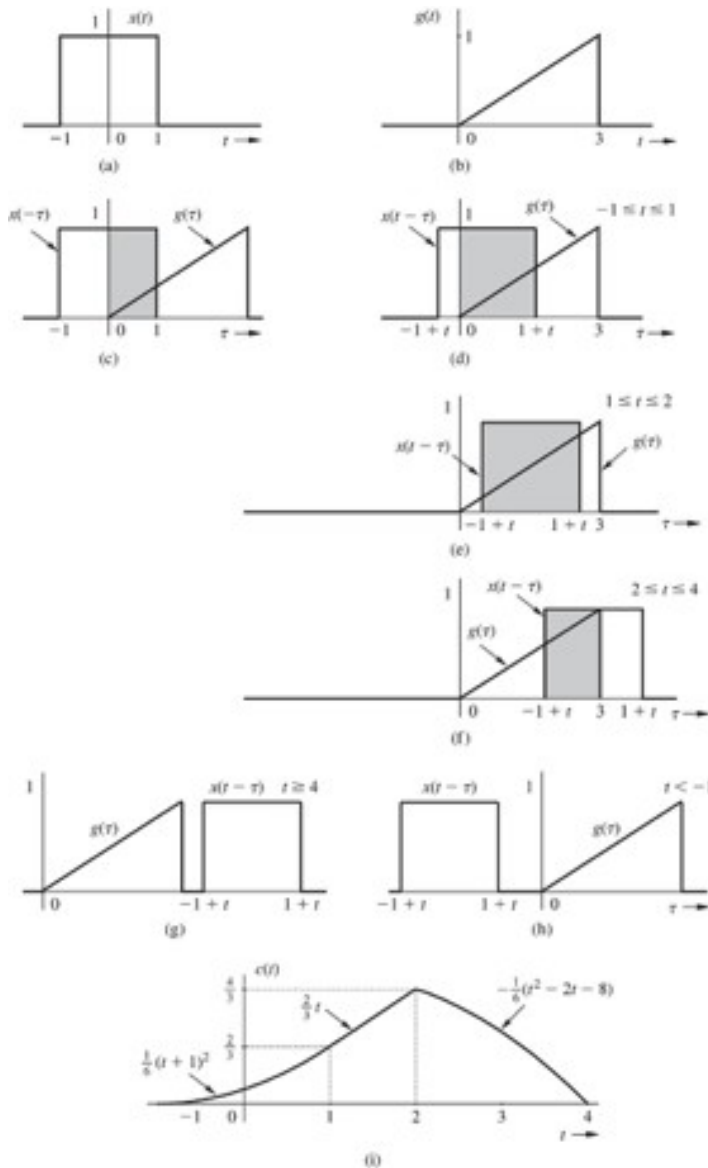
CT Convolution Graphical Approach



- Keep the function $f(t)$ fixed
- Flip $g(t)$ about $t=0$ to get $g(-t)$
- Shift $g(-t)$ along $f(t)$ by an amount t_0
- The area under the product of $f(t)$ and $g(t-t_0)$ is the value of the convolution at time t_0
- Remember convolution is commutative, so pick the easier signal to flip and slide.

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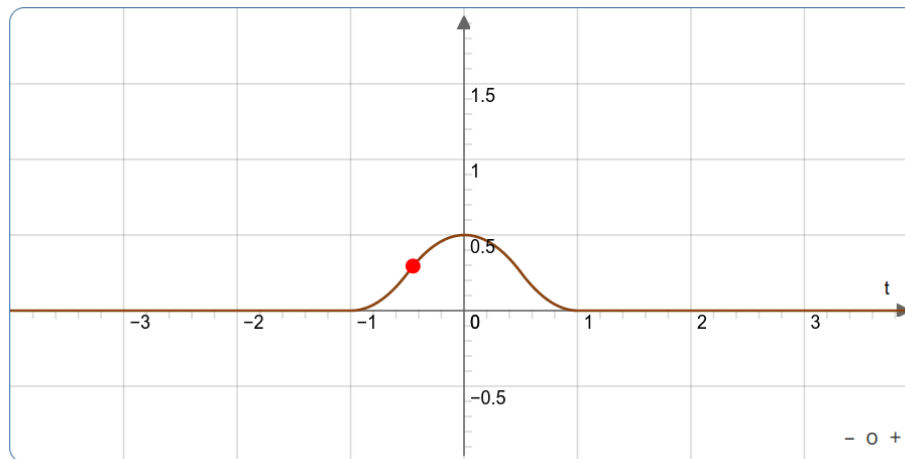
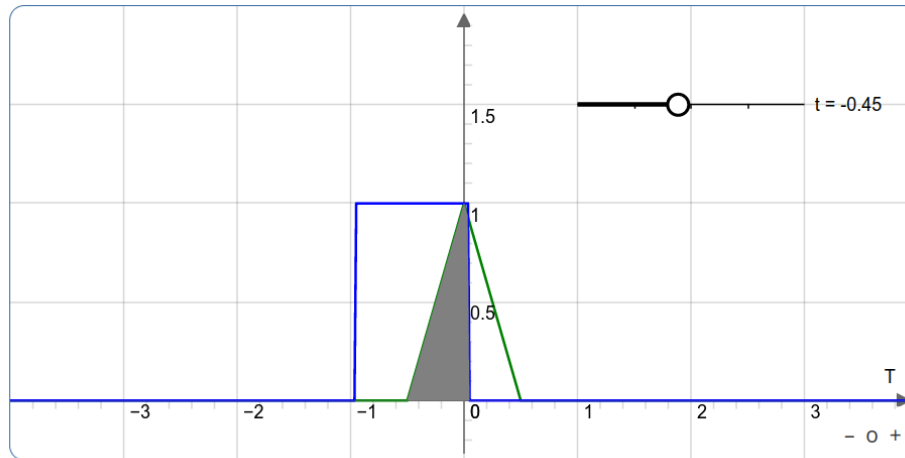
DT Convolution

$$(f * g)[n] = \sum_{m=-\infty}^{\infty} f[m]g[n - m]$$

Same process as CT:

- find $g[-n]$ and slide it along $f[n]$
- Multiply the aligned values and sum
- Start sliding it from a point where all calculated points are zero valued
- Keep going until you reach a point where all calculated points are zero valued
- The same convolution properties apply as in CT
- Keep track of the 'n' you are calculating!

Convolution Simulation



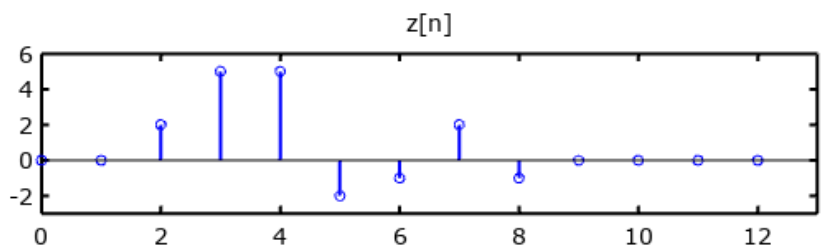
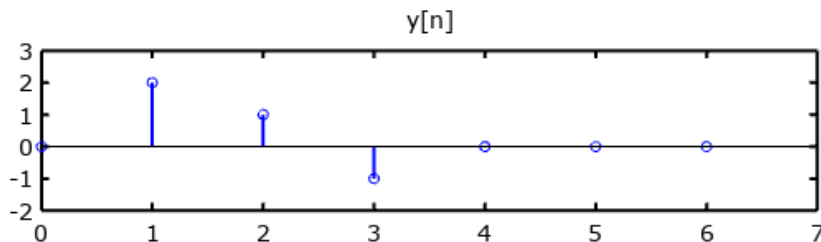
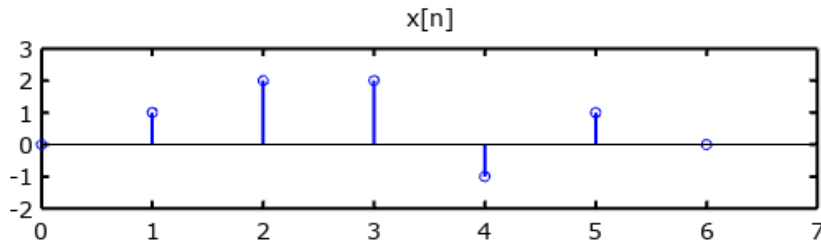
https://alexanderustyuzhanin.github.io/Web_Convolution/

DT Convolution

See W03A_DT_Convolution.m and try out some of your own $x[n]$ and $y[n]$'s

<https://octave.sourceforge.io/octave/function/conv.html>

<https://www.mathworks.com/help/matlab/ref/conv.html>



$$x[n] = \{0, 1, 2, 2, -1, 1, 0\}$$

$$y[n] = \{0, 2, 1, -1, 0, 0, 0\}$$

$$\text{for } n = \{0, 1, 2, 3, 4, 5, 6\}$$

$$z[n] = x[n] * y[n]$$

$$z[n] = \{0, 0, 2, 5, 5, -2, -1, 2, -1\}$$

$$\text{for } n = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$$

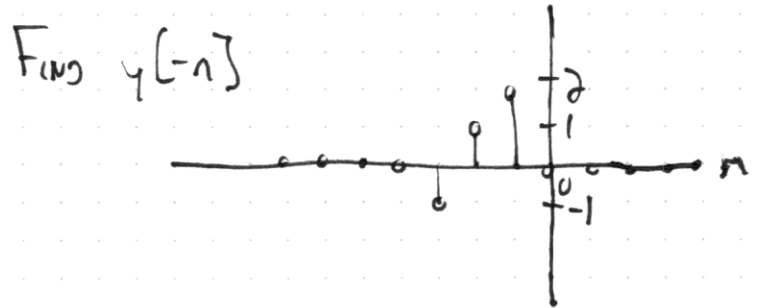
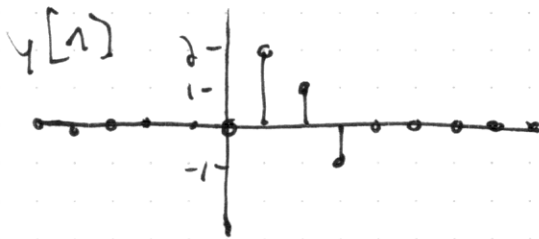
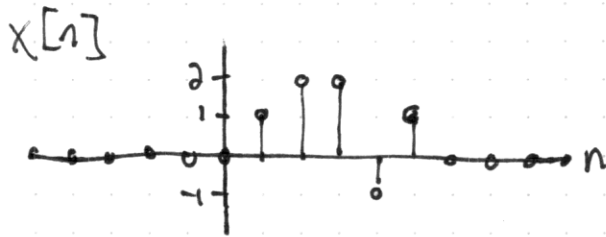
Don't lose track of the n 's - make sure we calculate the output at the right time!

DT CONVOLUTION

$$x[n] = \{0, 1, 2, 2, -1, 1, 0\}$$

$$y[n] = \{0, 2, 1, -1, 0, 0\}$$

$$n = \{0, 1, 2, 3, 4, 5, 6\}$$



$$z[n] = x[n] * y[n]$$

n	-3	-2	-1	0	1	2	3	4	5		
$x[n]$	0	0	0	0	1	2	2	-1	1	0	0
$y[-n]$	-1	1	2	0	0	0	0	0	0	$\Rightarrow$	0
$y[-n+1]$	0	-1	1	2	0	0	0	0	0	$\Rightarrow$	0
$y[-n+2]$	0	0	-1	1	2	0	0	0	0	$\Rightarrow$	$2 \times 1 = 2$
$y[-n+3]$	0	0	0	-1	1	2	0	0	0	$\Rightarrow$	$1 \times 1 + 2 \times 2 = 5$
$y[-n+4]$	0	0	0	0	-1	1	2	0	0	$\Rightarrow$	$1 \times (-1) + 1 \times 2 + 2 \times 2 = 5$
$y[-n+5]$	0	0	0	0	0	-1	1	2	0	$\Rightarrow$	$(-1) \times 2 + 1 \times 2 + 2 \times (-1) = -2$
$y[-n+6]$	0	0	0	0	0	0	-1	1	2	$\Rightarrow$	$-1 \times 2 + 1 \times (-1) + 2 \times 1 = -1$
$y[-n+7]$	0	0	0	0	0	0	0	-1	1	$\Rightarrow$	$-1 \times (-1) + 1 \times 1 = 2$
$y[-n+8]$	0	0	0	0	0	0	0	0	-1	$\Rightarrow$	$-1 \times 1 = -1$
$y[-n+9]$	0	0	0	0	0	0	0	0	0	$\Rightarrow$	0

$$z[n] = [0, 0, 2, 5, 5, -2, -1, 2, -1]$$

$$\text{For } n = [0, 1, 2, 3, 4, 5, 6, 7, 8]$$